American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science &Technology (FST)

SPRING 25-26

Section: D

Software Quality Assurance and Testing

PROJECT TITLE

AI-Based Harassment Detection and Complaint Management System

A Report submitted

By

SN	Student Name	Student ID
1	SHAMA ZERIN	22-49210-3
2	SUSMITA CHAKRABORTY TANNI	22-49208-3
3	MIRZA TARVIR BAIG	21-45611-3
4	ABIDA AFRIN	22-49212-3
5	MD. PARVEZ MOSHARROF	21-45635-3

Software Test Plan

For

AI-Based Harassment Detection and Complaint Management System

Version 1.0 approved

Prepared by SHAMA ZERIN, SUSMITA CHAKRABORTY TANNI, MIRZA TARVIR BAIG, ABIDA AFRIN, MD. PARVEZ MOSHARROF.

American International University-Bangladesh

19 May 2026

Table of Contents

Revision History

Revision	Date	Updated by	Update Comments
0.1	07.04.2026	Susmita Tanni	First Draft
0.2	11.04.2026	Shama Zerin	Second Draft
0.3	16.04.2026	Miraz Tarvir Baig	Updated functional and non-functional requirements verified through customer from UAT
0.4	20.04.2026	Abida Afrin	Third Draft
0.5	24.04.2026	Md. Parvez Mosharrof	Final Version

1. TEST PLAN IDENTIFIER: SQT-D04

2. REFERENCES

- Any reference documents with the test plan. For example: Software Requirement Specification (SRS) Document

3. INTRODUCTION

Background to the Problem

Harassment is a serious and widespread problem in schools, universities, workplaces, public spaces, and online platforms. Every day, many people face different forms of harassment such as verbal abuse, bullying, stalking, cyber harassment, blackmail, inappropriate messaging, and threatening behavior. However, a large number of victims do not report these incidents. Many people remain silent because they fear social humiliation, retaliation, lack of support, or the possibility that their complaint will not be taken seriously. As a result, many harassment cases remain unresolved, and offenders often continue their harmful behavior without consequences.

In many organizations and institutions, the complaint management process is still manual, unstructured, and time-consuming. Victims often need to report incidents through traditional methods such as paper forms, emails, or verbal communication, which can be confusing and emotionally distressing. In some cases, complaints are not documented properly, are delayed, or are handled without enough confidentiality. This creates a lack of trust in the complaint-handling system and discourages victims from seeking justice or support.

The problem becomes even more challenging in digital environments. Harassment now frequently occurs through social media, messaging platforms, online forums, emails, and digital communities. Because of the huge amount of online communication, it is difficult for administrators or authorities to manually identify harmful content quickly and accurately. This leads to delayed responses and allows harmful behavior to continue unchecked.

Another major issue is the difficulty of reviewing complaints fairly and efficiently. Some complaints may contain incomplete information, while others may require analysis of chat logs, screenshots, or text evidence. It becomes difficult for authorities to manually evaluate every complaint in a short time, especially when the number of complaints is high. Therefore, there is a growing need for a smart, secure, and structured system that can help users submit complaints safely, allow administrators to manage cases effectively, and use AI to detect harassment-related patterns more efficiently.

The harassment complaint management crisis is driven by several key factors:

• Fear of Reporting

Many victims hesitate to report harassment due to fear of social stigma, retaliation, embarrassment, or lack of action. Without a safe and confidential reporting system, many cases remain hidden.

• Lack of Structured Complaint Handling

Traditional complaint systems are often manual, slow, and poorly documented. This creates delays in investigation and reduces trust in the complaint resolution process.

• Increasing Digital Harassment

Online platforms have made harassment more frequent and difficult to monitor. Reviewing digital messages, comments, and reports manually takes too much time and effort.

• Privacy and Confidentiality Concerns

Harassment-related complaints often involve sensitive personal information. If complaint data is not handled securely, victims may face further risk and emotional harm.

• Difficulty in Case Analysis

Authorities often struggle to analyze complaints properly due to lack of evidence, incomplete information, or the large volume of reported incidents. This may delay or affect fair decision-making.

Challenges in Harassment Complaint Management

· Underreporting of Harassment Cases

A significant number of harassment incidents are never reported because victims feel unsafe, unsupported, or unsure about the complaint process. This allows many cases to go unnoticed.

· Delayed Response and Investigation

Manual complaint systems often cause delays in reviewing reports, verifying evidence, and taking necessary action. Such delays may worsen the victim's situation.

· Difficulty in Evidence Review

Harassment complaints may include screenshots, written statements, chat records, or digital files. Reviewing all these materials manually becomes difficult when case numbers increase.

- **Inconsistent Complaint Handling**

Without a structured system, different authorities may handle similar cases differently. This may create unfairness and reduce accountability.

- **Data Security Risks**

Sensitive complaint information must be stored securely. Weak security can result in unauthorized access, privacy violations, and loss of trust in the system.

Solution to the Problem

Objective:

The goal of this project is to develop a digital platform that allows users to report harassment incidents in a safe, structured, and confidential way. The system will connect victims, administrators, and complaint handlers through a secure complaint management process. By integrating AI-based text analysis and intelligent complaint handling, this platform aims to ensure that harassment complaints are reported, reviewed, and resolved more efficiently while maintaining fairness, privacy, and accountability.

Proposed Solutions:

- **Digital Complaint Submission System**

A secure online reporting system where users can submit harassment complaints with descriptions and supporting evidence. This will make the reporting process easier, faster, and better documented.

- **AI-Based Harassment Detection**

The system will use AI and natural language processing to analyze complaint text, messages, or written evidence in order to identify abusive, threatening, or inappropriate language patterns.

- **Role-Based Complaint Management**

A structured role-based system where users can submit and track complaints, while administrators can review, update, verify, and resolve reported cases. This ensures accountability and controlled access.

· **Complaint Prioritization and Smart Decision Support**

AI can help identify urgent or high-risk complaints by analyzing severity, repeated patterns, or offensive language. This helps authorities take faster and more informed action.

· **Secure Evidence Storage and Case Tracking**

All complaints, uploaded files, and case history will be stored securely so that every step of the complaint process can be tracked transparently from submission to resolution.

· **Privacy and Confidentiality Protection**

The system will apply strong authentication, authorization, and data protection measures so that complaint-related information remains safe and accessible only to authorized persons.

4. REQUEIREMNT SPECIFICATION

4.1 System Features

- User Registration & Authentication
- Complaint Submission with evidence upload
- AI-Based Harassment Detection using NLP
- Complaint Management & Tracking
- Notification System
- Report Generation & Analytics
- Secure Data Storage

Users (5 Types)

1. General User (Victim)
2. Admin
3. Moderator / Complaint Handler
4. Authority / Management
5. AI System (Automated Actor)

1. General User (Victim)

User Stories

SL	As a/an	I want to	So that	Acceptance
01	user	register	I can access the system	User can register with valid email& password;error shown for invalid input; account created successfully
02	user	Log in securely	my data remains protected	User can login with correct credentials;incorrect login shows error;password is encrypted
03	user	Submit a complaint	I can report harassment	Complain from submits successfully;required fields must be filled;conformation message shown
04	user	Upload evidence	My complaint is supported	User can upload files(image/pdf); file size/type validated;upload success message shown
05	user	Track complaint status	I know progress	User can view status (submitted/review/resolved); updates shown real-time
06	user	Receive notifications	I stay updated	User recievs notifications on status change;notifications visible in dashboard/email
07	user	Submit complaints anonymously	I feel safe	User can hide identity;system does not reveal personal info;complaint still submitted successfully
08	user	To view my complaint history	I can review past cases	User can see all previous complaints;details and status visible;sorted by date

2. Admin

User Stories

SL	As a/an	I want to	So that	Acceptance
01	Admin	Manage users	I can control access	Admin can view, add, edit, and remove user accounts; users can be assigned appropriate roles.
02	Admin	View all complaints	I can monitor the system	Admin can see a complete list of complaints; filter and

				search functionality available.
03	Admin	Assign complaints	They are handled properly	Admin can assign complaints to moderators/handlers; each complaint assigned to a responsible user.
04	Admin	Update complaint status	Users are informed	Admin can update complaint statuses (submitted/review/resolved); users are notified of status changes.
05	Admin	Generate reports	I can analyze trends	Admin can generate reports based on complaint data (e.g., number of complaints, types, severity).
06	Admin	Manage roles	Permissions are controlled	Admin can assign and modify roles for users (General User, Moderator, Authority).
07	Admin	Remove invalid complaints	System stays clean	Admin can delete complaints deemed invalid or duplicate; system reflects these changes immediately.
08	Admin	Monitor system performance	Issues are resolved quickly	Admin can view performance metrics (e.g., complaint resolution times, AI detection accuracy) and resolve issues promptly.

3. Moderator/Complaint Handler

User Stories

SL	As a/an	I want to	So that	Acceptance
01	Moderator	Review complaints	I can verify them	Moderator can view the complaint details and check for accuracy; invalid complaints flagged for further review.
02	Moderator	Analyze evidence	Decisions are accurate	Moderator can review uploaded evidence (images, chat logs, files) and

				make informed decisions based on available information.
03	Moderator	Categorize complaints	They are organized	Moderator can classify complaints into categories (e.g., verbal abuse, cyberbullying, stalking); categories visible for easy management.
04	Moderator	Update case progress	Tracking is clear	Moderator can update the status of a case (e.g., pending, in progress, resolved) and add notes; users are notified of updates.
05	Moderator	Communicate with users	I can request details	Moderator can send messages or requests for more information to users; communication is logged for transparency.
06	Moderator	Prioritize urgent complains	Critical cases are handled first	Moderator can prioritize complaints based on urgency (e.g., threats, repeated harassment); priority visible for faster handling.
07	Moderator	Mark cases resolved	Workflow completes	Moderator can mark complaints as resolved after completing the investigation and actions; system logs completion and notifies users.

08	Moderator	Flag suspicious complaints	Fraud is reduced	Moderator can flag complaints that seem fraudulent or suspicious for further investigation; flagged complaints are reviewed by admin or AI.

4. Authority/ Management

User Stories

SL	As a/an	I want to	So that	Acceptance
01	Authority	View reports	I can make decisions	Authority can generate and view detailed reports on complaints, trends, and system performance to inform decision-making.
02	Authority	Analyze trends	Policies can improve	Authority can analyze complaint data trends (e.g., frequency of incidents, common issues) to inform policy adjustments.
03	Authority	Monitor complaint statistics	Risks are indentified	Authority can view statistics (e.g., types of complaints, resolution times) to identify potential risks or issues within the system.
04	Authority	Review serious cases	Proper action is taken	Authority can review high-priority cases and ensure that appropriate actions are taken quickly and fairly.

05	Authority	Ensure compliance	Rules are followed	Authority can monitor compliance with internal policies and legal regulations to ensure fairness in handling complaints.
06	Authority	Audit complaint handling	Fairness is maintained	Authority can audit the complaint handling process to ensure fairness, transparency, and efficiency.
07	Authority	Identify repeat offender's	Preventive action is taken	Authority can identify repeat offenders based on complaint history and implement preventive actions (e.g., warnings, bans).
08	Authority	Improve policies	Harassments reduce	Authority can modify policies based on insights from the system, aiming to reduce harassment incidents over time.

5. AI System (Automated Actor)

User stories

SL	As a/an	I want to	So that	Acceptance
01	AI system	Analyze complaint text	Harassment is detected	AI system analyzes complaint text and detects patterns indicative of harassment (e.g., abusive language, threats).

02	AI system	Classify severity	Urgent cases are prioritized	AI system classifies the severity of complaints (e.g., low, medium, high) to help prioritize urgent cases for faster resolution.
03	AI system	Detect abusive language	Moderation is easier	AI system identifies abusive, threatening, or inappropriate language within complaints to assist moderators in reviewing cases more effectively.
04	AI system	Learn from past data	Accuracy improves	AI system improves its analysis accuracy over time by learning from past complaint data, refining its detection and classification.
05	AI system	Flag high risk cases	Admin act quickly	AI system flags high-risk complaints (e.g., threats or repeated harassment) for immediate attention from admins or moderators.
06	AI system	Assist moderators	Decision making is faster	AI system provides recommendations or insights to moderators, speeding up the decision-making process.
07	AI system	Generate insights	Trends are visible	AI system generates insights from complaint data, helping authorities and admins identify patterns or emerging issues.

08	AI system	Reduce false positives	Fairness is maintained	AI system continuously improves its accuracy to reduce false positives, ensuring that complaints are handled fairly.

List down the system functional requirements that describes the system's functionalities:

1. User Registration Functional Requirements

1.1 The system shall allow new users to register using valid personal information such as full name, email, phone number, gender, and password.

1.2 The system shall validate the email format and phone number before account creation.

1.3 The system shall ensure that the password meets security requirements (minimum length, special character, uppercase letter, etc.).

1.4 The system shall prevent duplicate account registration using the same email address.

1.5 The system shall encrypt and securely store user passwords in the database.

1.6 The system shall allow users to add emergency contact information during registration.

1.7 The system shall display confirmation messages after successful registration.

Priority Level: High
Precondition: User has internet access and valid registration information.

2. System Login Functional Requirements

2.1 The software shall allow users to login with their given username and password

2.2 If the username and/or password has been inserted wrong for more than three times, the random verification code will be generated by the system to retry login.

2.3 If the number of login attempt exceed its limit (5 times), the system shall block the user account login for one hour *[optional function]*

Priority Level: High
Precondition: user have valid user id and password

2.4 The system shall maintain secure authenticated sessions after successful login.

2.5 The system shall allow users to logout securely from the system.

Priority Level: High
Precondition: User must have a registered account.

3. Complaint Submission Functional Requirements

3.1 The system shall allow users to submit harassment complaints digitally.

3.2 The system shall provide a complaint form containing title, description, category, incident date, and evidence upload options.

3.3 The system shall validate required complaint fields before submission.

3.4 The system shall allow users to submit complaints anonymously.

3.5 The system shall generate a unique complaint ID for each submitted complaint.

3.6 The system shall store complaint data securely in the database.

3.7 The system shall display a confirmation message after successful complaint submission.

Priority Level: High
Precondition: User must be logged into the system.

4. Evidence Upload Functional Requirements

4.1 The system shall allow users to upload evidence files such as images, PDFs, screenshots, audio, or chat records.

4.2 The system shall validate file type and file size before upload.

4.3 The system shall securely store uploaded evidence linked with the complaint ID.

4.4 The system shall prevent malicious or unsupported files from being uploaded.

4.5 The system shall allow moderators and admins to access uploaded evidence for review purposes.

Priority Level: High
Precondition: Complaint submission form is active

5. AI-Based Harassment Detection Functional Requirements

5.1 The system shall analyze complaint text using AI and NLP techniques.

5.2 The system shall detect abusive, threatening, or inappropriate language patterns.

5.3 The system shall classify complaint severity levels such as Low, Medium, and High.

5.4 The system shall flag high-risk complaints for immediate review.

5.5 The AI system shall provide recommendation support to moderators during complaint evaluation.

5.6 The system shall continuously improve detection accuracy using historical complaint data.

Priority Level: High
Precondition: Complaint text must be submitted successfully.

6. Complaint Tracking Functional Requirements

6.1 The system shall allow users to track complaint status in real time.

6.2 The system shall display complaint stages such as Submitted, Under Review, In Progress, and Resolved.

6.3 The system shall maintain complaint history for each user account.

6.4 The system shall allow users to view complaint details and previous updates.

6.5 The system shall notify users whenever complaint status changes.

Priority Level: Medium
Precondition: Complaint must exist in the system.

7. Notification System Functional Requirements

7.1 The system shall send notifications for complaint status updates.

7.2 The system shall notify users through dashboard alerts and email notifications.

7.3 The system shall notify moderators when new complaints are assigned.

7.4 The system shall send alerts for high-priority harassment cases.

7.5 The system shall maintain notification history for users.

Priority Level: Medium
Precondition: Notification service must be enabled.

8. Admin Management Functional Requirements

8.1 The system shall allow admins to view all registered users.

8.2 The system shall allow admins to add, edit, block, or remove user accounts.

8.3 The system shall allow admins to assign complaints to moderators.

8.4 The system shall allow admins to update complaint statuses.

8.5 The system shall allow admins to remove duplicate or invalid complaints.

8.6 The system shall allow admins to generate complaint reports and analytics.

8.7 The system shall allow admins to manage user roles and permissions.

Priority Level: High
Precondition: Admin must have valid administrative access.

9. Moderator Functional Requirements

9.1 The system shall allow moderators to review complaint details and evidence.

9.2 The system shall allow moderators to categorize complaints based on harassment type.

9.3 The system shall allow moderators to update complaint progress and investigation notes.

9.4 The system shall allow moderators to communicate with users for additional information.

9.5 The system shall allow moderators to prioritize urgent complaints.

9.6 The system shall allow moderators to mark complaints as resolved after investigation.

9.7 The system shall allow moderators to flag suspicious or fraudulent complaints.

Priority Level: High
Precondition: Moderator must be assigned authorized complaints.

10. Report Generation Functional Requirements

10.1 The system shall generate complaint reports based on complaint type, severity, and resolution status.

10.2 The system shall allow admins and authorities to export reports in PDF or Excel format.

10.3 The system shall provide graphical analytics and complaint statistics.

10.4 The system shall allow authorities to analyze complaint trends and repeated offenders.

10.5 The system shall generate monthly and yearly complaint summaries.

Priority Level: Medium
Precondition: Complaint data must exist in the database.

11. Security and Privacy Functional Requirements

11.1 The system shall ensure secure authentication and authorization for all users.

11.2 The system shall encrypt sensitive complaint data and personal information.

11.3 The system shall restrict unauthorized access to complaint information.

11.4 The system shall maintain audit logs for complaint handling activities.

11.5 The system shall protect anonymous complainant identities from unauthorized disclosure.

11.6 The system shall automatically logout inactive users after a predefined inactivity period.

Priority Level: High
Precondition: System security services must be active.

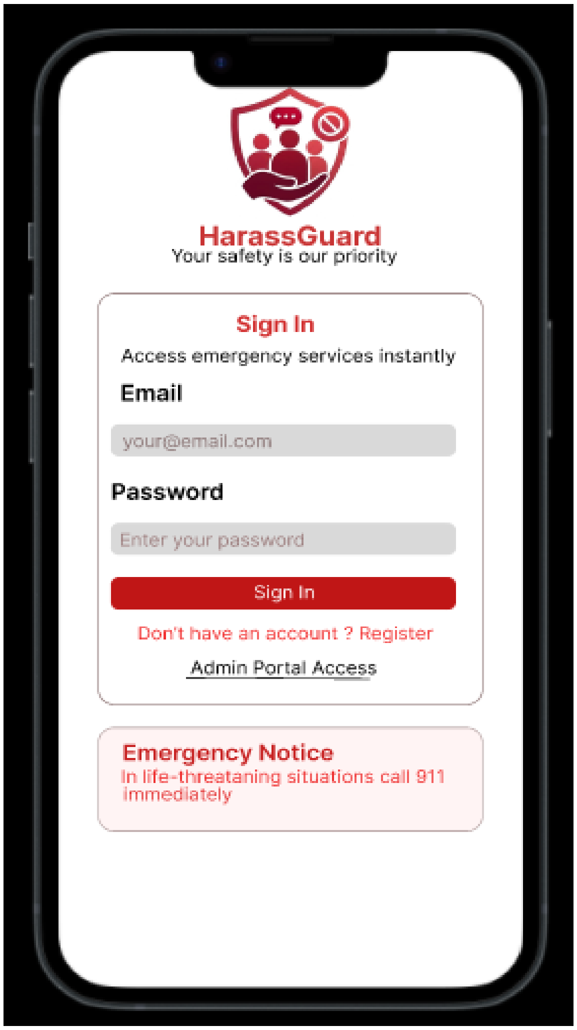
4.2 System Quality Attributes :

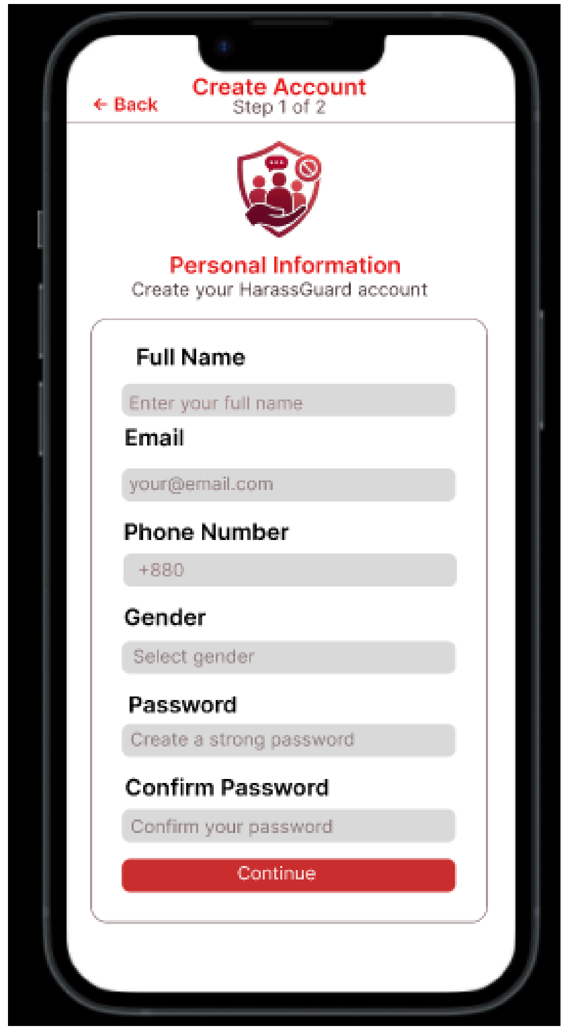
Serial	***Criteria***	**Description**
1	Usability	A first-time user shall be able to complete and submit a harassment complaint with evidence in under 5 minutes without requiring external assistance.

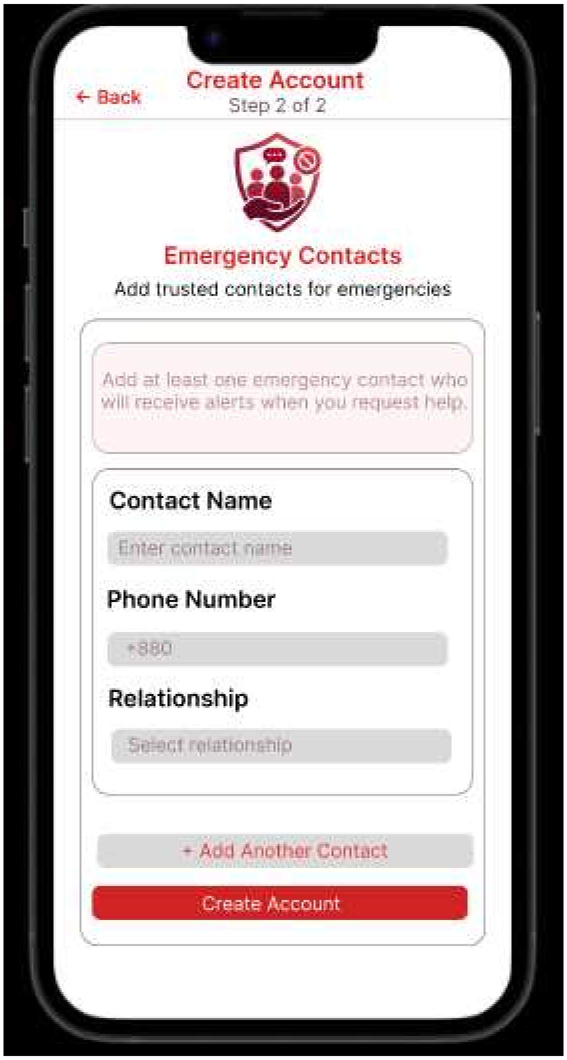
2	Confidentiality	Complaint data and victim identities must be encrypted; personal information shall only be accessible to authorized personnel (Admin/Moderator) and must never be revealed in anonymous submissions.
3	Reliability	The system must maintain 99.9% availability to ensure victims can report incidents 24/7, especially in time-sensitive or threatening situations.
4	AI Accuracy	The AI system shall achieve a minimum of 85% accuracy in detecting abusive language patterns and high-risk cases to reduce false positives and ensure fair handling.
5	Responsiveness	Automated notifications for complaint status updates must be delivered to the user's dashboard or email within 60 seconds of a status change.
6	Integrity	The system must prevent any unauthorized modification of chat logs or uploaded evidence to maintain the legal and investigative value of the data.
7	Scalability	The system shall be capable of handling a 50% increase in concurrent users and submitted complaints without a degradation in page load times exceeding 3 seconds.

4.3 System Interface

- Draw the system interface where the users will interact with the system's functionality.


HarassGuard
Your safety is our priority
Sign In
Access emergency services instantly
Email
your@email.com
Password
Enter your password
Sign In
Don't have an account ? Register
Admin Portal Access
Emergency Notice
In life-threataning situations call 911 immediately


← Back
Create Account
Step 1 of 2
Personal Information
Create your HarassGuard account
Full Name
Enter your full name
Email
your@email.com
Phone Number
+880
Gender
Select gender
Password
Create a strong password
Confirm Password
Confirm your password
Continue



4.4 Project Requirements

Budget Estimation

Cost Category	Estimated Cost(BDT)	Details
Human Resources (Developers, Testers, PM)	2,25,000	5 members × 6 months × 7,500 avg.
Infrastructure & Server (Cloud Hosting)	35,000	AWS/Azure for 8 months
Third-Party APIs (AI/NLP, Email, Storage)	30,000	OpenAI API, SendGrid, AWS S3

Testing Tools & Licenses	20,000	Selenium, JMeter, Postman Pro
Development Tools & Software	20,000	IDEs, DB tools, design software
Documentation & Miscellaneous	10,000	Printing, meetings, contingency
TOTAL	**340000**	**Approved budget ceiling**

Time Estimation

Phase	**Duration**	**Key Activities**
Requirements & Planning	1 month	Stakeholder interviews, SRS, use-case modeling
System Design	1 month	Architecture, DB schema, UI wireframes
Development – Module 1 (Auth, Complaint Submission)	1 month	Login, registration, complaint form, evidence upload
Development – Module 2 (AI Detection, NLP)	1.5 months	AI model integration, abusive text detection, prioritization
Development – Module 3 (Admin Dashboard, Reports)	1 month	Admin panel, case management, analytics, notifications
Testing (Unit, Integration, Acceptance)	1.5 months	Test design, execution, defect fixing
Deployment & Training	1 month	Production deploy, user training, handover
TOTAL	**8 month**	**Estimated completion**

5. FEATURES NOT TO BE TESTED

The following areas will not be specifically or directly addressed in the test plan for the **AI-Based Harassment Detection and Complaint Management System**. Any testing coverage in these areas will be incidental, arising as a by-product of other testing activities.

- **Third-Party AI/NLP Library Internal Training and Accuracy** The system uses pre-trained NLP models (e.g., transformers, spaCy) for harassment detection. The internal training data, model architecture, and inherent accuracy of these third-party libraries are outside the scope of this project. Testing will only verify that the AI module correctly returns a classification (harassment type, severity) when given an input text; the underlying model's research-level performance will not be tested.

- **Cloud Infrastructure and Hosting Platform** The system is deployed on a cloud provider (e.g., AWS, Azure, or Google Cloud). The reliability, scalability, or security patches of the underlying cloud infrastructure are the responsibility of the provider. Testing will assume a stable cloud environment; only the application's behaviour under normal conditions will be verified.

- **Email and SMS Delivery Infrastructure** Notification messages (status updates, alerts) are sent via third-party email and SMS gateways (e.g., SendGrid, Twilio). The delivery success rate, latency, or carrier-level filtering of these gateways is not tested. The system will be tested to confirm that the notification trigger is correctly invoked; actual receipt at the user's device is the responsibility of the third-party service.

- **Client-Side Browser Security and Device Hardware** The behaviour of the web application across different browser versions (Chrome, Firefox, Edge) or device screen sizes is tested for basic compatibility, but deep OS-level security (e.g., cookie handling, local storage encryption) or device-specific hardware issues are not within scope. Users' personal devices and their security configurations are the responsibility of the end user.

- **External Database Backup and Disaster Recovery** The system uses a database (e.g., PostgreSQL, MySQL). Automated backup, restore procedures, and disaster recovery plans are managed by the IT operations team and are not part of the functional test plan. Testing will only verify that data is correctly stored and retrieved during normal operations.

- **Manual/Offline Complaint Handling** Any complaint submitted through paper forms, emails, or verbal communication outside the digital system is completely out of scope. The test plan only covers complaints submitted via the online complaint submission module.

- **Third-Party Authentication Providers (if integrated later)**
  If future versions integrate OAuth (Google, Facebook) or institutional SSO, those providers

Internal authentication mechanisms will not be tested. The system will only test the receipt of success/failure callbacks from those providers.

- **Administrator's Own Reporting Tools** If administrators export complaint data (CSV/JSON) and process it using external tools (Excel, Power BI, Tableau), those external applications are under the customer's control. Testing of customer-created dashboards or analyses is the responsibility of the system administrator.

6. TESTING APPROACH

6.1 Testing Levels

The testing for the AI-Based Harassment Detection and Complaint Management System will consist of three primary test levels: Unit Testing**,** System/Integration Testing (combined), and Acceptance Testing. Given the academic project constraints and team size, the Test Manager will lead the majority of testing activities, with active participation from all four team members. Selenium WebDriver will be used as the main automation tool for system/integration testing of the web interface.

- **Unit Testing**
  - Unit Testing will be conducted by the individual developer responsible for each module:
    - User registration & authentication
    - Complaint submission & file upload
    - AI harassment detection engine (with mock AI outputs for unit tests)
    - Complaint management & status tracking
    - Notification system
    - Admin/moderator dashboards
    - Report generation & analytics
  - Each developer must produce proof of unit testing — including a test case list, sample output logs, data printouts, and any defect information — before the unit is approved by the team leader.
  - All unit test documentation will be handed over to the Test Manager for reference during integration testing.
  - No unit will be passed to System/Integration testing if it has any Critical defect outstanding (e.g., system crash, data loss, complete failure of a core user story).
- **System / Integration Testing (Combined)**
  - System/Integration Testing will be performed by the Test Manager and Development Team Leade**r**, with assistance from individual developers as required.
  - Selenium WebDriver will be used to automate the testing of end-to-end workflows through the web browser. Selenium scripts will simulate real user actions (clicking, typing, uploading files, navigating pages) and verify expected UI, error messages, and state changes.
  - A module will enter System/Integration testing only after all Critical defects from unit testing have been resolved.
  - A module may carry up to two Major defects into integration testing, provided those defects do not block testing of other features (i.e., a workaround exists, such as manually resetting a state).

- Integration testing will focus on verifying the interactions between modules via the user interface:
  - Complaint submission → AI detection → severity classification → database storage
  - Admin assigns complaint → Moderator receives notification → Moderator updates status → User receives notification
  - File upload → secure storage → retrieval during case review
  - Role-based access control (General User, Admin, Moderator, Authority)
- Integration test cases will be derived directly from the User Stories defined for each of the five actor types (General User, Admin, Moderator, Authority, AI System).
- Selenium test scripts will be written for the most critical and repetitive workflows (e.g., login, complaint submission, status updates) and run on multiple browsers (Chrome, Firefox) using browser drivers. Test results (pass/fail, screenshots on failure) will be logged automatically.

- **Acceptance Testing**
  - Acceptance Testing will be performed by representative end users — e.g., students/faculty as victims, a volunteer moderator, an admin — with the assistance of the Test Manager and Development Team Leader.
  - Acceptance testing will run in parallel with a simulated manual complaint process for a period of two weeks after completion of System/Integration Testing. During this parallel run, the same set of sample complaints will be processed both manually (paper/email) and through the system; results will be compared to validate correctness and efficiency.
  - Acceptance test scenarios will be derived directly from the Acceptance Criteria (AC) defined in the Requirement Specification section for each user story (e.g., AC01–AC08 for General User, AC01–AC08 for Admin, etc.).
  - Acceptance is granted only when:
    - All High-priority user stories pass their Acceptance Criteria without any Critical or Major defects.
    - The AI harassment detection shows ≥80% agreement with human moderator judgment on a test set of 50 labeled complaints.
    - Users are able to complete the core flow (register → submit complaint → track status → receive resolution) without confusion.

6.2 Test Tools

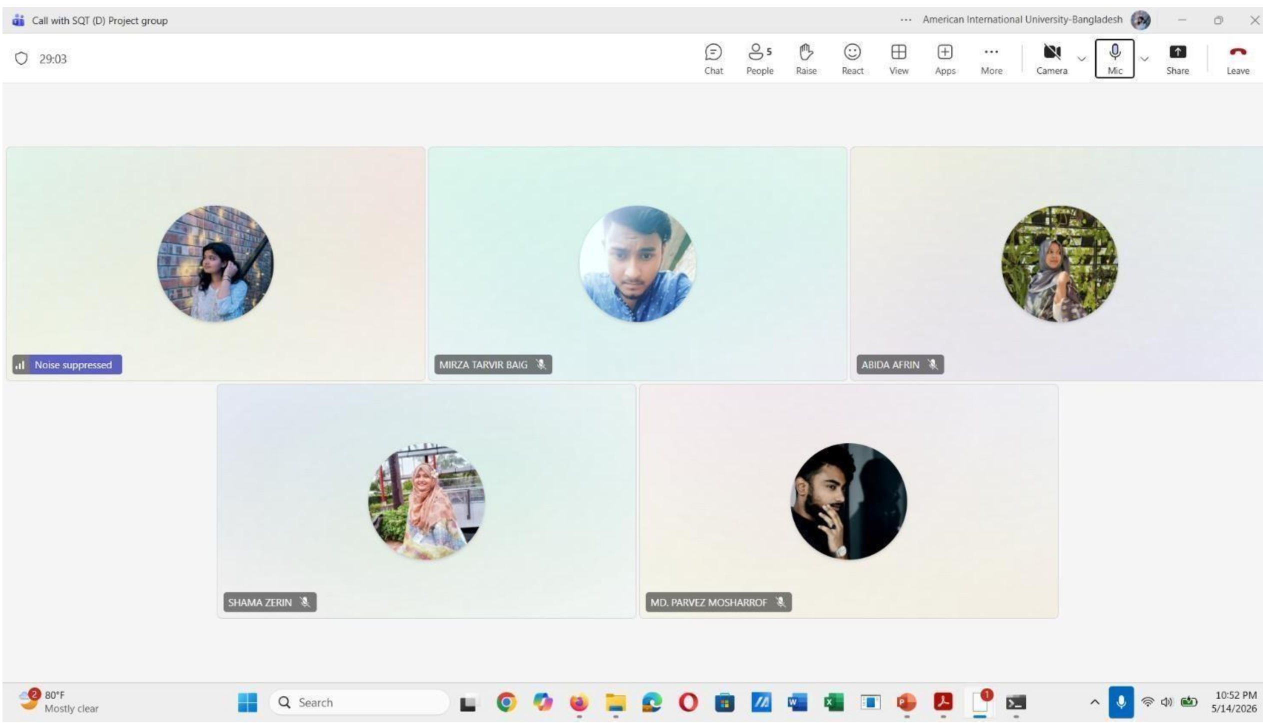
For this project, Selenium WebDriver will be used for automated browser testing, supported by manual exploratory testing and lightweight utilities. No API testing tools (e.g., Postman) will be used; all backend verification will be performed indirectly through the UI and database inspection.

- **1. Automated Functional Testing with Selenium WebDriver**
  - Selenium scripts will automate the following critical user journeys:
    - User registration with valid/invalid data and duplicate email handling
    - Login with correct/wrong credentials and account lockout simulation
    - Complaint submission (required fields, evidence upload, anonymous option)
    - Complaint tracking and status display

        - Admin dashboard: user management, complaint assignment, report generation
        - Moderator dashboard: reviewing complaints, updating status, adding notes
    - Scripts will be written in Python using Selenium WebDriver.
    - Test execution will be performed on Chrome and Firefox browsers.
    - Assertions will verify:
        - Presence and correctness of UI elements (text, buttons, error messages)
        - Navigation flows and page redirects
        - File upload success/failure messages
        - Role-based access (e.g., normal user cannot see admin panels)
    - On test failure, Selenium will capture a screenshot automatically.
    - No headless browser execution is required; tests will be run visibly on the team's local machines.
- **2. Manual Functional & Exploratory Testing**
    - In addition to Selenium automation, testers will manually execute test cases that are difficult to automate (e.g., complex UI interactions, visual layout checks, real-time notification delivery).
    - Manual testing will also cover edge cases and negative scenarios not yet scripted.
    - Each manual test case will be documented in a Google Sheets test log with columns: Test Case ID, Description, Steps, Expected Result, Actual Result, Status (Pass/Fail), Screenshot link.
    - The team will perform exploratory testing during the acceptance phase to uncover unexpected behaviour (e.g., slow page loads, broken back-button navigation, confusing error messages).
- **3. Bug Tracking**
    - All defects found (by Selenium scripts or manually) will be logged in a Google Sheets bug tracker with the following columns:
        - Bug ID, Module, Steps to Reproduce, Severity (Critical/Major/Minor/Trivial), Status (Open/Fixed/Retest/Closed), Assigned To, Selenium-related (Yes/No)
    - Selenium test reports will be attached as evidence.
- **4. Test Data Management**
    - A set of 50 sample complaint texts (labelled by the team as "harassment" / "not harassment" / "threat" / "cyberbullying") will be used for AI validation.
    - Sample evidence files (images, PDFs, dummy chat logs) will be prepared in advance and stored in a shared /test-data folder.
    - Selenium scripts will use these files for upload tests.
- **5. Version Control & Documentation**
    - GitHub repository will store:
        - Selenium test scripts (e.g., /tests/selenium/)
        - Test data files
        - Test logs and bug tracker (as CSV or Markdown)
        - This test plan document
    - All test artifacts will be versioned alongside the application source code.
- **6. Browser & Driver Setup**
    - ChromeDriver and GeckoDriver (for Firefox) will be downloaded and configured.
    - Browser versions will be kept stable during the testing period to avoid environment inconsistencies.

6.3 Meetings

Meeting Date	Signature				
26.3.2026	Tanni	Shama	Abida	Mirza	Parvez
3.4.2026	Tanni	Shama	Abida	Mirza	Parvez
9.4.2026	Tanni	Shama	Abida	Mirza	Parvez
15.4.2026	Tanni	Shama	Abida	Mirza	Parvez
23-4-2026	Tanni	Shama	Abida	Mirza	Parvez



7. TEST CASES/TEST ITEMS

Project Name: HarassGuard	Test Designed by: Shama Zerin
Test Case ID: FR_01	Test Designed date: 01/03/26
Test Priority (Low, Medium, High): High	Test Executed by: Shama Zerin
Module Name: User Registration	Test Execution date: 01/04/26
Test Title: Verify account registration with valid personal information	
Description: Test new user registration with full name, email, phone, gender and password	
Precondition (If any): User must not already have an account	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the HarassGuard website 2. Click on Sign Up 3. Enter Full Name 4. Enter Email address 5. Enter Phone Number 6. Enter Gender 7. Enter Password and Confirm Password 8. Click Continue	Name: Rita Akter Email: rita@gmail.com Phone: +8801712345678 Gender: Female Password: Rita@123	User should be able to proceed to Step 2 of account creation	As expected	Pass
Post Condition: User personal information is stored and account creation moves to Step 2.				

Project Name: HarassGuard	Test Designed by: Shama Zerin
Test Case ID: FR_02	Test Designed date: 01/03/26
Test Priority (Low, Medium, High): High	Test Executed by: Shama Zerin
Module Name: Emergency Contact Setup	Test Execution date: 02/04/26
Test Title: Verify emergency contact addition during account creation	
Description: Test the emergency contact page during account registration	
Precondition (If any): User has completed Step 1 of registration (personal information)	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Complete personal information page 2. Navigate to Emergency Contact page 3. Enter Contact Name 4. Enter Contact Phone Number 5. Enter Relationship 6. Tap Add Another Contact to add a second contact 7. Click Create Account	Contact 1: Antara, 01632345678, Sister Contact 2: Arnab, 01734567890, Brother	Both emergency contacts should be saved and account created successfully	As expected	Pass
Post Condition: Emergency contacts are stored in the database and account is created.				

Project Name: HarassGuard	Test Designed by: Shama Zerin
Test Case ID: FR_03	Test Designed date: 02/03/26
Test Priority (Low, Medium, High): High	Test Executed by: Shama Zerin
Module Name: User Login	Test Execution date: 03/04/26
Test Title: Verify login with valid email and password	

Description: Test the login page with correct credentials
Precondition (If any): User must have a registered account

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the HarassGuard website 2. Click on Login 3. Enter registered email address 4. Enter password 5. Click Login button	Email: rita@gmail.com Password: Rita@123	User should be redirected to the Home page successfully	As expected	Pass
Post Condition: User session is created and user is redirected to the dashboard/home page.				

Project Name: HarassGuard	Test Designed by: Shama Zerin
Test Case ID: FR_04	Test Designed date: 02/03/26
Test Priority (Low, Medium, High): High	Test Executed by: Shama Zerin
Module Name: Home Page Navigation	Test Execution date: 04/04/26
Test Title: Verify Home Page loads correctly with all UI elements	
Description: Test that the home page displays all feature modules and navigation icons	
Precondition (If any): User must be logged into the HarassGuard application	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to the application 2. Observe the Home page loads 3. Verify SOS Emergency button is visible 4. Verify feature modules are present: Report Incident, Safe Zone, Contact Access, AI Detector, Complaint Tracking 5. Verify bottom navigation icons: Home, Alert, Feedback, Settings	Valid login credentials Home page URL	Home page should load with all modules, SOS button and navigation icons visible	as expected	pass
Post Condition: Home page is fully loaded and all UI elements are accessible.				

Project Name: HarassGuard	Test Designed by: Susmita Chakraborty Tanni
Test Case ID: FR_05	Test Designed date: 03/03/26
Test Priority (Low, Medium, High): High	Test Executed by: Susmita Chakraborty Tanni
Module Name: SOS Emergency - Single Tap	Test Execution date: 05/04/26
Test Title: Verify SOS button single tap starts voice recording	

Description: Test that a single tap on the SOS Emergency button initiates voice recording
Precondition (If any): User must be logged in and on the Home page; microphone permission must be granted

Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Login to the application 2. Navigate to Home page 3. Locate the SOS Emergency button 4. Single tap the SOS Emergency button 5. Observe system response	SOS Button Single tap gesture	Voice recording should start immediately after single tap	As expected	Pass
Post Condition: Voice recording begins and is stored securely in the system.				

Project Name: HarassGuard	Test Designed by: Susmita Chakraborty Tanni
Test Case ID: FR_06	Test Designed date: 03/03/26
Test Priority (Low, Medium, High): High	Test Executed by: Susmita Chakraborty Tanni
Module Name: SOS Emergency - Double Tap	Test Execution date: 05/04/26
Test Title: Verify SOS button double tap redirects to Emergency type selection	
Description: Test that a double tap on the SOS button navigates to the Emergency type selection screen	
Precondition (If any): User must be logged in and on the Home page	

Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Login to the application 2. Navigate to Home page 3. Locate the SOS Emergency button 4. Double tap the SOS Emergency button 5. Observe screen navigation	SOS Button Double tap gesture	System should redirect user to Emergency type selection screen	As expected	Pass
Post Condition: User is navigated to the Emergency type selection screen.				

Project Name: HarassGuard	Test Designed by: Susmita Chakraborty Tanni
Test Case ID: FR_07	Test Designed date: 04/03/26
Test Priority (Low, Medium, High): High	Test Executed by: Susmita Chakraborty Tanni
Module Name: Complaint Submission	Test Execution date: 07/04/26
Test Title: Verify complaint submission with valid information	
Description: Test the Report Incident feature for submitting a harassment complaint	
Precondition (If any): User must be logged in and required fields must be filled	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to application 2. Tap Report Incident from Home page 3. Enter complaint description 4. Select harassment type/category 5. Click Submit button 6. Observe confirmation message	Description: I was verbally abused by a colleague on 15/05/26 Category: Verbal Abuse	Complaint should be submitted successfully and confirmation message should be shown	As expected	Pass
Post Condition: Complaint is stored in the database and assigned a unique complaint ID.				

Project Name: HarassGuard	Test Designed by: Susmita Chakraborty Tanni
Test Case ID: FR_08	Test Designed date: 05/03/26
Test Priority (Low, Medium, High): High	Test Executed by: Susmita Chakraborty Tanni
Module Name: Evidence Upload	Test Execution date: 08/04/26
Test Title: Verify evidence file upload during complaint submission	
Description: Test that users can upload image or PDF evidence with a complaint	
Precondition (If any): User must be on the complaint submission page	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to application 2. Tap Report Incident 3. Fill complaint details 4. Tap Upload Evidence button 5. Select an image file from device 6. Click Submit button	File: screenshot.jpg File type: image/jpeg File size: 1.2 MB	File should upload successfully and upload success message should appear	As expected	Pass
Post Condition: Evidence file is stored securely and linked to the complaint.				

Project Name: HarassGuard	Test Designed by: Mirza Tarvir Baig
Test Case ID: FR_09	Test Designed date: 05/03/26
Test Priority (Low, Medium, High): Medium	Test Executed by: Mirza Tarvir Baig
Module Name: Anonymous Complaint	Test Execution date: 09/04/26
Test Title: Verify anonymous complaint submission	
Description: Test that users can submit a complaint without revealing personal identity	
Precondition (If any): User must be logged in and on the complaint submission page	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to application 2. Tap Report Incident 3. Enable anonymous toggle or checkbox 4. Enter complaint description 5. Click Submit button 6. Observe complaint submission result	Anonymous: Yes Description: Threatening messages received via social media	Complaint should be submitted successfully without exposing user identity	As expected	Pass
Post Condition: Complaint is stored without personal identifiers; user identity is protected.				

Project Name: HarassGuard	Test Designed by: Mirza Tarvir Baig
Test Case ID: FR_10	Test Designed date: 06/03/26
Test Priority (Low, Medium, High): Medium	Test Executed by: Mirza Tarvir Baig
Module Name: Complaint Tracking	Test Execution date: 10/04/26
Test Title: Verify complaint status tracking	
Description: Test that users can view the current status of their submitted complaints	
Precondition (If any): User must have at least one previously submitted complaint	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to application 2. Tap Complaint Tracking from Home page 3. View list of submitted complaints 4. Select a specific complaint 5. Observe status information	Complaint ID: HG-00123 Status: Under Review	Complaint status should be displayed correctly (Submitted / Under Review / Resolved)	As expected	Pass
Post Condition: Complaint status is displayed accurately in real time.				

Project Name: HarassGuard	Test Designed by: Mirza Tarvir Baig
Test Case ID: FR_11	Test Designed date: 07/03/26
Test Priority (Low, Medium, High): Medium	Test Executed by: Mirza Tarvir Baig
Module Name: AI Detector	Test Execution date: 11/04/26
Test Title: Verify AI-based harassment detection on submitted text	
Description: Test that the AI Detector correctly identifies harassment in input text	
Precondition (If any): User must be logged in and AI Detector feature must be active	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to application 2. Tap AI Detector from Home page 3. Enter a sample text message 4. Click Analyze button 5. Observe AI detection result	Input: You are worthless and I will make your life miserable	AI should detect the text as harassment and display the result with severity level	Expenses did not categorize correctly	Fail
Post Condition: Detection result is logged and displayed to the user with severity classification.				

Project Name: HarassGuard	Test Designed by: Mirza Tarvir Baig
Test Case ID: FR_12	Test Designed date: 07/03/26
Test Priority (Low, Medium, High): High	Test Executed by: Mirza Tarvir Baig
Module Name: Notification System	Test Execution date: 12/04/26
Test Title: Verify notification received on complaint status change	
Description: Test that users receive a notification when their complaint status is updated	
Precondition (If any): User must have a submitted complaint and notifications must be enabled	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Admin or moderator updates a complaint status to Resolved 2. Login to the HarassGuard application as the complainant 3. Check the notification panel 4. Open the notification 5. Observe notification content	Complaint ID: HG-00123 Status updated to: Resolved	User should receive an in-app notification about the status change	As expected	Pass
Post Condition: Notification is delivered and marked as read after user views it.				

Project Name: HarassGuard	Test Designed by: Abida Afrin
Test Case ID: FR_13	Test Designed date: 08/03/26
Test Priority (Low, Medium, High): Medium	Test Executed by: Abida Afrin
Module Name: Safe Zone	Test Execution date: 13/04/26
Test Title: Verify Safe Zone feature loads and displays location correctly	
Description: Test that the Safe Zone feature correctly shows user location and nearby safe places	
Precondition (If any): User must be logged in; location permission must be granted	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to application 2. Tap Safe Zone from Home page 3. Allow location access if prompted 4. Observe map and safe zone markers 5. Tap a marker to view details	Location: Dhaka, Bangladesh Location permission: Granted	Safe Zone map should load and display nearby safe locations correctly	As expected	Pass
Post Condition: Safe zone locations are displayed correctly based on user's current location.				

Project Name: HarassGuard	Test Designed by: Abida Afrin
Test Case ID: FR_14	Test Designed date: 09/03/26
Test Priority (Low, Medium, High): Medium	Test Executed by: Abida Afrin
Module Name: Contact Access	Test Execution date: 14/04/26
Test Title: Verify Contact Access page displays emergency contacts	
Description: Test that the Contact Access feature shows the user's saved emergency contacts	
Precondition (If any): User must be logged in and have emergency contacts saved during registration	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to the application 2. Tap Contact Access from Home page 3. View the list of emergency contacts 4. Tap a contact to call or message	Emergency Contact 1: Antara, Sister Emergency Contact 2: Arnab, Brother	All saved emergency contacts should be displayed with name, phone and relationship	As expected	Pass
Post Condition: Emergency contacts are displayed correctly and are accessible for quick communication.				

Project Name: HarassGuard	Test Designed by: Abida Afrin
Test Case ID: FR_15	Test Designed date: 09/03/26
Test Priority (Low, Medium, High): Medium	Test Executed by: Abida Afrin
Module Name: Profile Settings	Test Execution date: 15/04/26
Test Title: Verify user profile update via side menu	
Description: Test that users can update profile information through the side menu	
Precondition (If any): User must be logged in	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to the application 2. Tap the side menu icon from top-left corner 3. Tap Profile option 4. Update phone number or address 5. Tap Save button	New Phone: +8801898765432 Address: Mirpur, Dhaka	Updated profile information should be saved successfully	As expected	Pass
Post Condition: Updated profile information is stored in the database.				

Project Name: HarassGuard	Test Designed by: Abida Afrin
Test Case ID: FR_16	Test Designed date: 10/03/26
Test Priority (Low, Medium, High): Medium	Test Executed by: Abida Afrin
Module Name: Password Reset	Test Execution date: 16/04/26
Test Title: Verify password reset via OTP verification	
Description: Test the forgot password functionality using OTP	
Precondition (If any): User must have a registered account with a valid email	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the login page 2. Click Forgot Password 3. Enter registered email address 4. Enter OTP code received via email 5. Enter new password 6. Confirm new password 7. Click Reset Password	Email: rita@gmail.com OTP: 847231 New Password: Rita@456	Password should be updated successfully after OTP verification	As expected	Pass
Post Condition: Password is updated in the database and user can login with new password.				

Project Name: HarassGuard	Test Designed by: Md. Parvez Mosharrof
Test Case ID: FR_17	Test Designed date: 10/03/26
Test Priority (Low, Medium, High): Medium	Test Executed by: Md. Parvez Mosharrof
Module Name: Feedback	Test Execution date: 17/04/26
Test Title: Verify feedback submission through bottom navigation	
Description: Test that users can submit feedback via the Feedback section in bottom navigation	
Precondition (If any): User must be logged in	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to application	Subject: Feature Suggestion	Feedback should be submitted	As expected	Pass

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
2. Tap Feedback icon from bottom navigation bar 3. Enter feedback subject 4. Type feedback message 5. Tap Submit button 6. Observe confirmation message	Message: Please add a panic button for quick police alert	successfully and a confirmation message should appear		
Post Condition: Feedback is stored in the system and confirmation message is shown to user.				

Project Name: HarassGuard	Test Designed by: Md. Parvez Mosharrof
Test Case ID: FR_18	Test Designed date: 11/03/26
Test Priority (Low, Medium, High): High	Test Executed by: Md. Parvez Mosharrof
Module Name: Admin - Complaint Assignment	Test Execution date: 18/04/26
Test Title: Verify admin can assign complaint to a moderator	
Description: Test that admin can assign a submitted complaint to a moderator for handling	
Precondition (If any): Admin must be logged in; at least one unassigned complaint must exist	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Admin 2. Open the Complaints list from Admin dashboard 3. Select an unassigned complaint 4. Click Assign button 5. Select a moderator from dropdown 6. Click Confirm Assignment	Complaint ID: HG-00124 Assigned Moderator: Parvez	Complaint should be assigned to the selected moderator and status updated accordingly	As expected	Pass
Post Condition: Complaint is assigned and moderator receives a notification about the new case.				

Project Name: HarassGuard	Test Designed by: Md. Parvez Mosharrof
Test Case ID: FR_19	Test Designed date: 12/03/26
Test Priority (Low, Medium, High): Medium	Test Executed by: Md. Parvez Mosharrof
Module Name: Moderator - Case Resolution	Test Execution date: 19/04/26
Test Title: Verify moderator can mark a complaint as resolved	
Description: Test that a moderator can update and close a complaint after investigation	
Precondition (If any): Moderator must be logged in; an assigned complaint must be available	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Moderator 2. Open assigned complaint from dashboard 3. Review complaint details and evidence 4. Add investigation notes 5. Click Mark as Resolved button 6. Confirm resolution	Complaint ID: HG-00123 Notes: Offender warned and case documented	Complaint status should change to Resolved and user should be notified	Not as expected	fail
Post Condition: Complaint is marked as resolved, logged in the system, and complainant is notified.				

Project Name: HarassGuard	Test Designed by: Md. Parvez Mosharrof
Test Case ID: FR_20	Test Designed date: 13/03/26
Test Priority (Low, Medium, High): High	Test Executed by: Md. Parvez Mosharrof
Module Name: Complaint History	Test Execution date: 20/04/26
Test Title: Verify complaint history display for logged-in user	
Description: Test that users can view all previously submitted complaints with correct details	
Precondition (If any): User must be logged in and must have submitted at least one complaint	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login to application 2. Tap the side menu icon from top-left corner 3. Tap My Complaints or Complaint History option 4. View list of past complaints 5. Select a complaint to view full details	User: rita@gmail.com Previous Complaint ID: HG-00120	All previous complaints should be listed with ID, date, type and current status	As expected	Pass
Post Condition: Complaint history is displayed correctly and sorted by submission date.				

8. ITEM PASS/FAIL CRITERIA

The test process will be completed once all assigned testers have executed their respective test cases and the test results have been reviewed by the Test Manager. The system will be considered ready for deployment only when all High priority test cases pass without any Critical defects. A maximum of two Medium priority test cases may remain with Minor defects, provided that documented workarounds are available.

Test Case ID	Test Title	Status (Pass/Fail)
FR_01	Verify account registration with valid personal information	Pass
FR_02	Verify emergency contact addition during account creation	Pass
FR_03	Verify login with valid email and password	Pass
FR_04	Verify Home Page loads correctly with all UI elements	Pass
FR_05	Verify SOS button single tap starts voice recording	Pass
FR_06	Verify SOS button double tap redirects to Emergency type selection	Pass
FR_07	Verify complaint submission with valid information	Pass
FR_08	Verify evidence file upload during complaint submission	Pass
FR_09	Verify anonymous complaint submission	Pass
FR_10	Verify complaint status tracking	Pass
FR_11	Verify AI-based harassment detection on submitted text	Fail
FR_12	Verify notification received on complaint status change	Pass
FR_13	Verify Safe Zone feature loads and displays location correctly	Pass
FR_14	Verify Contact Access page displays emergency contacts	Pass
FR_15	Verify user profile update via side menu	Pass
FR_16	Verify password reset via OTP verification	Pass
FR_17	Verify feedback submission through bottom navigation	Pass
FR_18	Verify admin can assign complaint to a moderator	Pass
FR_19	Verify moderator can mark a complaint as resolved	Fail
FR_20	Verify complaint history display for logged-in user	Pass

9. STAFFING AND TRAINING NEEDS

Team Member Name & ID	Assigned Testing Role	Required Training (before System/Integration Testing)
SHAMA ZERIN (22-49210-3)	**Test Manager** – Overall test planning, defect tracking, acceptance testing coordination	Test plan creation, defect management (Jira/Sheet), ethical handling of complaint data
SUSMITA CHAKRABORTY TANNI (22-49208-3)	**QA Tester (Full-time)** – Manual functional testing, execution of test cases, AI output validation	Selenium testing, NLP basics for harassment detection (precision/recall), test case documentation
MIRZA TARVIR BAIG (21-45611-3)	**Development Team Leader** – Unit testing oversight, integration testing participation, defect fixing	Regression testing strategy, Selenium, secure file upload testing

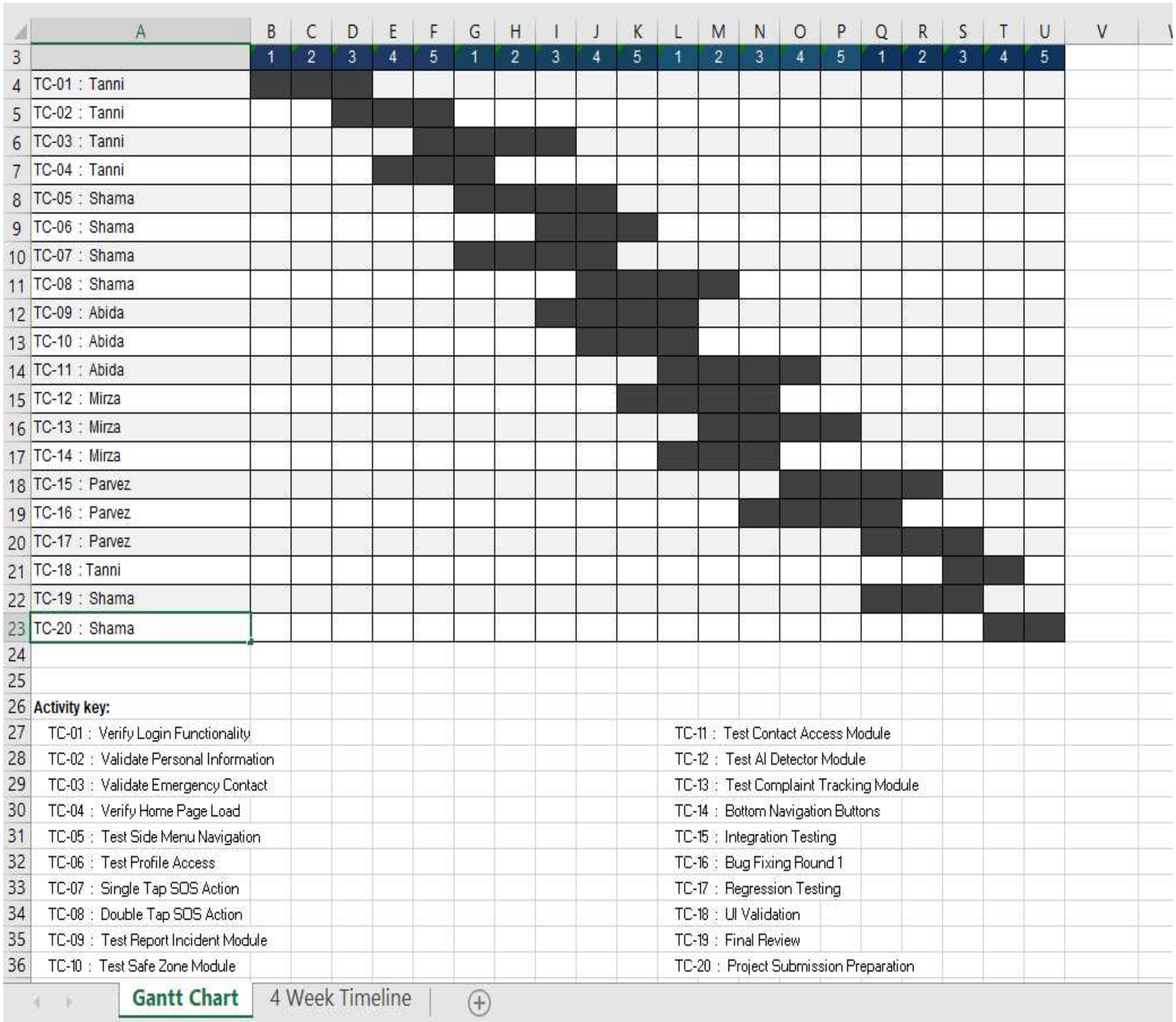
ABIDA AFRIN (22-49212-3)	**Test Analyst** – Test data design (harassment text samples), Selenium testing, results recording	Creating test data sets, Selenium collections, boundary value analysis for complaint forms
MD. PARVEZ MOSHARROF (21-45635-3)	**Security & Compliance Tester** – Role-based access control (RBAC), privacy testing (anonymous complaints), secure storage validation	RBAC testing techniques, data encryption basics, OWASP top 10 for web apps

10. RESPONSIBILITIES

	TM	PM	Dev Team	Test Team	Client
Acceptance test Documentation & Execution	X	X		X	X
System/Integration test Documentation & Exec.	X		X	X	
Unit test documentation & execution	X		X	X	
System Design Reviews	X	X	X	X	X
Detail Design Reviews	X	X	X	X	
Test procedures and rules	X	X	X	X	
Screen & Report prototype reviews			X	X	X
Change Control and regression testing	X	X	X	X	X

11. TESTING SCHEDULE

Week 1 — April 5 to April 9, 2026								
TC-01	Login	Verify Login Functionality	Week 1	Day 1	05-Apr-2026	07-Apr-2026	3	Completed
TC-02	Create Account	Validate Personal Information	Week 1	Day 3	07-Apr-2026	09-Apr-2026	3	Completed
TC-03	Create Account	Validate Emergency Contact	Week 1	Day 5	09-Apr-2026	12-Apr-2026	4	Completed
TC-04	Home Page	Verify Home Page Load	Week 1	Day 4	08-Apr-2026	10-Apr-2026	3	Completed
Week 2 — April 10 to April 14, 2026								
TC-05	Home Page	Test Side Menu Navigation	Week 2	Day 1	10-Apr-2026	13-Apr-2026	4	Completed
TC-06	Home Page	Test Profile Access	Week 2	Day 3	12-Apr-2026	14-Apr-2026	3	Completed
TC-07	SOS	Single Tap SOS Action	Week 2	Day 1	10-Apr-2026	13-Apr-2026	4	Completed
TC-08	SOS	Double Tap SOS Action	Week 2	Day 4	13-Apr-2026	16-Apr-2026	4	Completed
TC-09	Modules	Test Report Incident Module	Week 2	Day 3	12-Apr-2026	15-Apr-2026	4	Completed
TC-10	Modules	Test Safe Zone Module	Week 2	Day 4	13-Apr-2026	15-Apr-2026	3	Completed
TC-12	Modules	Test AI Detector Module	Week 2	Day 5	14-Apr-2026	17-Apr-2026	4	Completed
Week 3 — April 15 to April 19, 2026								
TC-11	Modules	Test Contact Access Module	Week 3	Day 1	15-Apr-2026	18-Apr-2026	4	Completed
TC-13	Modules	Test Complaint Tracking Module	Week 3	Day 2	16-Apr-2026	19-Apr-2026	4	Completed
TC-14	Navigation	Bottom Navigation Buttons	Week 3	Day 1	15-Apr-2026	17-Apr-2026	3	Completed
TC-15	System	Integration Testing	Week 3	Day 4	18-Apr-2026	21-Apr-2026	4	Completed
TC-16	System	Bug Fixing Round 1	Week 3	Day 3	17-Apr-2026	20-Apr-2026	4	Completed
Week 4 — April 20 to April 24, 2026								
TC-17	System	Regression Testing	Week 4	Day 1	20-Apr-2026	22-Apr-2026	3	Completed
TC-18	System	UI Validation	Week 4	Day 3	22-Apr-2026	23-Apr-2026	2	Completed
TC-19	System	Final Review	Week 4	Day 1	20-Apr-2026	22-Apr-2026	3	Completed
TC-20	Project	Project Submission Preparation	Week 4	Day 4	23-Apr-2026	24-Apr-2026	2	Completed



12. PLANNING RISKS AND CONTINGENCIES

Risk Description	Probability	Impact	Mitigation Plan
AI Model Accuracy & Bias: The NLP mode may fail to achieve 85% accuracy, show bias against certain language patterns, or produce high false positives/negatives leading to unfai complaint handling.	High	Critical	- Test with a diverse, pre-labelled dataset of 50+ complaints covering various harassment types. - Implement a human-in- the-loop review where moderators can override AI classifications. - Plan for continuous retraining of the model

			with validated complaint data post-launch.
Data Privacy & Security Breach: Sensitive personal information (victim identity, complaint details, evidence) could be exposed due to SQL injection, broken access control, or insufficient encryption, violating confidentiality requirements.	Moderate	Critical	- Enforce strict Role- Based Access Control (RBAC) testing for all 5 user types. - Conduct OWASP Top 10 security tests (SQLi XSS) before deployment. - Ensure all sensitive data is encrypted at rest and in transit (TLS 1.3). - Anonymize complainant identity in the database for anonymous submissions.
Evidence File Upload Vulnerabilities: Users may upload malicious files (e.g., scripts, executables) disguised as evidence, leading to server compromise or data corruption.	Moderate	High	- Implement strict server- side file type validation (allow only images, PDFs, .txt). - Scan all uploads for malware. - Store files outside the web root and serve them via a secure, authenticated handler.
System Unavailability (Downtime): The system may experience downtime, preventing victims from reporting incidents, especially during time-sensitive emergencies, violating the 99.9% availability requirement.	Low	Critical	- Deploy on a reliable cloud platform (AWS/Azure) with auto- scaling. - Implement database redundancy and regular automated backups. - Monitor system health with uptime alerts.
Poor AI Responsiveness: The AI detection module may take >3 seconds to analyze complaint text, slowing down the submission	Moderate	High	- Optimize AI model inference by using a lighter faster model or caching. - Set a timeout for AI calls; if exceeded, submit

process and creating a poor user experience during a stressful situation.			the complaint without AI analysis and trigger a background job. - Load test the API endpoint for concurrent AI analysis requests.
Notification Delivery Failure: Email or in-app notifications for complaint status updates may fail to reach users leading to lack of trust and missed updates, violating the responsiveness quality attribute.	Moderate	Moderate	- Implement a retry mechanism for failed email deliveries. - Log all notification attempts. - Provide a manual "refresh status" option in the user dashboard as a fallback.
Incomplete or Changing Requirements: New features or changes requested late in the testing cycle may impact the schedule and introduce new defects.	Moderate	Moderate	- Freeze requirements after the System Design phase. - Use a formal change request process; any approved change will trigger regression testing of affected modules. - Prioritize core functionalities (complaint submission, AI detection admin assignment) for initial release.
Third-Party API Dependency (Email/SMS): The system relies on external email/SMS gateways (e.g., SendGrid, Twilio). These servicescould experience downtime or rate-limiting, affecting notification delivery.	Low	Moderate	- Abstract the notification service behind an interface to allow switching providers. - Implement queuing for notifications so they are resent if the initial attempt fails. - Have a backup email provider option.
Team Availability & Skill Gaps: Team members may be	Moderate	Moderate	- Cross-train team members on Selenium and manual test case

unavailable due to other academic commitments or lack sufficient expertise in Selenium automation or NLP testing.			execution. - Maintain a shared, up-to- date test log and bug tracker in Google Sheets. - Schedule regular weekly sync meetings to identify blockers early.
Browser/Device Incompatibility: The web application may not render or function correctly on older browsers or specific mobile devices, limiting accessibility for some users.	Low	Minor	- Run Selenium test scripts on Chrome and Firefox during system testing. - Perform manual exploratory testing on at least 2 different mobile screen sizes (e.g., iPhone and Android). - Use responsive design frameworks (e.g., Bootstrap) to ensure basic compatibility.

13. APROVALS

Project Sponsor - Steve Sponsor	
Development Management - Ron Manager	
EDI Project Manager - Peggy Project	
RS Test Manager - Dale Tester	
RS Development Team Manager - Dale Tester	
Reassigned Sales - Cathy Sales	
Order Entry EDI Team Manager - Julie Order	